

16. Abstract

Pedestrian and bicycle crashes have been over-represented in comparison with their share in commute and all trips share because they are vulnerable road users and usually exposed and unprotected to the traffic. The fatality rates of pedestrians and bicyclists have been significantly higher than national average rates in Washington, DC. Improving the safety features of pedestrians and bicyclists in the nation's capital is a high priority. The previously conducted studies of pedestrian and bicycle crashes did not include in-detail analysis.

In this study, three years of pedestrian and bicycle crashes (2012-14) were digitized, classified, and analyzed. The classification was based on NHTSA PBCAT and recently proposed LMCM crash typologies. Majority of crashes happened at intersections (68%) and drivers were at fault or violation. The main pedestrian crash type was vehicle left turns (parallel paths) and of bicycles was the case of open door to traffic. The NHTSA crash groups were also identified and top three groups were examined in details and applicable PEDSAFE and BIKESAFE countermeasures were also listed. Due to few cases for some original NHTSA crash groups, new pedestrian and bicycle crash groups were proposed to categorize crashes better in Washington, DC area. Decision trees were developed using CHAID method to investigate contributing factors in fatal and severe injury (disabling) crashes. Traffic control type, crash time, alcohol, speeding, light condition, road type, city quadrant, and fault were contributing factors in more severe pedestrian crashes. Bicycle-only crashes and crashes at construction zones were the identified factors attributing in more severe bicycle crashes. The shortcomings of police crash report forms that were used were also discussed.